INCH-POUND
MIL-M-38510/123B
23 March 2004
SUPERSEDING
MIL-M-38510/123A
21 September 1982

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, CMOS, NEGATIVE LOGIC, ANALOG SWITCH, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Reactivated for new design as of 23 March 2004. May be used for either new or existing design acquisition.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF-38535

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS negative logic analog switches. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number. For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535, (see 6.3)

1.2 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-38535, and as specified herein.

1.2.1 Device types. The device types are as follows:

Device type	Circuit
01	Dual SPST switch
02	Quad SPST switch

NOTE: A channel is defined as a driver with associated switches.

1.2.2 Device class. The device class is the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outline. The case outline are as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
C	GDIP1-T14 or CDIP2-T14	14	Dual-in-line
E	GDIP1-T16 or CDIP2-T16	16	Flat pack
I	MACY1-X10	10	Can

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1.3 Absolute maximum ratings. 1/

Supply voltage between V+ and V-:	
Device types 01, 02	+40 V
Digital input voltage	$(+V_{supply}) \geq V_{IN} \geq (-0.3\ V)$
Analog input voltage	$(+V_{supply} +2\ V) \geq V_A \geq (-V_{supply} -2\ V)$
Storage temperature	-65°C to +150°C
Lead temperature (soldering, 10 seconds)	+300°C
Junction temperature (T_J)	175°C
Maximum current:	
Any terminal (except S or D)	30 mA
S or D terminals	20 mA

1.4 Recommended operating conditions.

Positive supply voltage	+15 V dc
Negative supply voltage	- 15 V dc
Reference voltage	Open
Input (address) high voltage	2.4 V
V_{IL}	0.8 V
Ambient operating temeperature range (T_A)	-55°C to +125°C

1.5 Power and thermal characteristics.

Case outline	Maximum allowable power dissipation 2/	Maximum θ_{JC}	Maximum θ_{JA}
C, E	400 mW @ T_A = 125°C	35°C/W	120°C/W
I	350 mW @ T_A = 125°C	40°C/W	140°C/W

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard for Microelectronics.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines.

(Copies of these documents are available online at http://assist.daps.dla.mil;quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

1/ Stresses above the absolute maximum rating may cause permanent damage to the device. Extended operation at the maximum levels may degrade performance and affect reliability.
2/ All leads welded or soldered to PC board.

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.4).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Terminal connections. The circuit diagram and terminal connections shall be as specified on figure 1.

3.3.2 Truth table. The truth table shall be as specified on figure 2.

3.3.3 Schematic circuits. The schematic circuits shall be maintained by the manufacturer and made available to the qualifying activity and the preparing activity (DSCC-VA) upon request.

3.3.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.6 Electrical test requirements. Electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.8 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.9 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 82 (see MIL-PRF-38535, appendix A).

4. VERIFICATION.

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer’s Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein.

4.2 Screening. Screening shall be in accordance with MIL-PRF-38535, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Additional screening for space level product shall be as specified in MIL-PRF-38535.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions V+ = 15 V, GND = 0 V, V- = -15 V Unless otherwise specified	Temperature range	Device type	Limits		Unit
					Min	Max	
Switch “ON” resistance (figure 3)	R_{DS}	V_{IN} = 0.8 V, V_S = 10 V, I_D = -1 mA	T_A = -55°C, +25°C	01		70	Ω
			T_A = 125°C			100	
			T_A = -55°C, +25°C	02		175	
			T_A = 125°C			250	
		V+ = 10 V, V- = -10 V, V_{IN} = 0.8 V , V_S = 7.5 V, I_D = -1 mA	T_A = -55°C, +25°C	01		100	
			T_A = 125°C			150	
			T_A = -55°C, +25°C	02		200	
			T_A = 125°C			250	
		V_{IN} = 0.8 V, V_S = -10 V, I_D = 1 mA	T_A = -55°C, +25°C	01		70	
			T_A = 125°C			100	
			T_A = -55°C, +25°C	02		175	
			T_A = 125°C			250	
		V+ = 10 V, V- = -10 V, V_{IN} = 0.8 V , V_S = -7.5 V, I_D = 1 mA	T_A = -55°C, +25°C	01		100	
			T_A = 125°C			150	
			T_A = -55°C, +25°C	02		200	
			T_A = 125°C			250	
Source “OFF” leakage current (figure 4)	$I_{S(OFF)}$	V_S = 14 V, V_D = -14 V, V_{IN} = 2.4 V	T_A = 25°C	01,02	-2	2	nA
			T_A = 125°C		-100	100	
			T_A = -55°C		-100	100	
		V_S = -14 V, V_D = 14 V, V_{IN} = 2.4 V	T_A = 25°C		-2	2	
			T_A = 125°C		-100	100	
			T_A = -55°C		-100	100	
Drain “OFF” leakage current (figure 5)	$I_{D(OFF)}$	V_D = -14 V, V_S = 14 V, V_{IN} = 2.4 V	T_A = 25°C	01,02	-2	2	nA
			T_A = 125°C		-100	100	
			T_A = -55°C		-100	100	
		V_D = 14 V, V_S = -14 V, V_{IN} = 2.4 V	T_A = 25°C		-2	2	
			T_A = 125°C		-100	100	
			T_A = -55°C		-100	100	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions V+ = 15 V, GND = 0 V, V- = -15 V Unless otherwise specified	Temperature range	Device type	Limits		Unit
					Min	Max	
Channel "ON" leakage current (figure 6)	$I_{D(ON)}$	$V_D = V_S = 14$ V, $V_{IN} = 0.8$ V	$T_A = 25°C$	01	-2	2	nA
				02	-2	2	
			$T_A = 125°C$	01,02	-200	200	
			$T_A = -55°C$		-200	200	
		$V_D = V_S = -14$ V, $V_{IN} = 0.8$ V	$T_A = 25°C$	01	-2	2	
				02	-2	2	
			$T_A = 125°C$	01,02	-200	200	
			$T_A = -55°C$		-200	200	
Low level input current (figure 7)	I_{IL} 1/	$V_{IL} = 0.8$ V, $V_{IH} = 2.4$ V	$T_A = 25°C$	01,02	-0.5	0.5	µA
			$T_A = 125°C$		-1.0	1.0	
			$T_A = -55°C$		-1.0	1.0	
High level input current (figure 7)	I_{IH}	$V_{IL} = 0.8$ V, $V_{IN} = 15$ V	$T_A = 25°C$	01,02	-0.5	0.5	µA
			$T_A = 125°C$		-1.0	1.0	
			$T_A = -55°C$		-1.0	1.0	
Supply current (figure 8)	$+I_{CC}$	$V_{IL} = 0$ V, (all inputs)	$T_A = 25°C$, 125°C	01,02		1.5	mA
			$T_A = -55°C$			2.0	
		$V_{IH} = 5$ V, (all inputs)	$T_A = 25°C$, 125°C			1.5	
			$T_A = -55°C$			2.0	
Supply current (figure 8)	$-I_{CC}$	$V_{IL} = 0$ V, (all inputs)	$T_A = 25°C$, 125°C	01,02	-1.5		mA
			$T_A = -55°C$		-2.0		
		$V_{IH} = 5$ V, (all inputs)	$T_A = 25°C$, 125°C		-1.5		
			$T_A = -55°C$		-2.0		
Capacitance: address	C_A	$T_A = 25°C$, GND = 0 V, f = 1 MHz, (see 4.4.1c), $V_{IL} = 0$ V		01,02		15	pF
Capacitance: Input switch	C_{IS}	$T_A = 25°C$, GND = 0 V, f = 1 MHz, (see 4.4.1c), $V_{IH} = 5$ V		01,02		15	pF

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions V+ = 15 V, GND = 0 V, V- = -15 V Unless otherwise specified	Temperature range	Device type	Limits		Unit
					Min	Max	
Capacitance: Output switch	C_{OS}	T_A = 25°C, GND = 0 V, f = 1 MHz, (see 4.4.1c), V_{IH} = 5 V		01,02		20	pF
Off isolation	V_{ISO}	T_A = 25°C, f = 200 kHz, V_{gen} = 1 V_{p-p}, (see 4.4.1d, 4.4.3c, and 4.4.4)		01,02	60		dB
Crosstalk between channels	V_{CT}	T_A = 25°C, f = 200 kHz, V_{gen} = 1 V_{p-p}, (see 4.4.1d, 4.4.3c, and 4.4.4)		01,02	60		dB
Charge transfer error	V_{CTE}	T_A = 25°C, (see 4.4.1d, 4.4.3c, and 4.4.4)		01,02	-10	10	mV
Turn “ON” time	$t_{(ON)}$	C_L = 100 pF, R_L = 1 kΩ, (Figure 9)	T_A = 125°C	01,02		800	ns
			T_A = -55°C, 25°C			600	
Turn “OFF” time	$t_{(OFF)}$	C_L = 100 pF, R_L = 1 kΩ, (Figure 9)	T_A = 125°C	01,02		650	ns
			T_A = -55°C, 25°C			500	

1/ Input current at one input node.

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TABLE II. Electrical test requirements.

MIL-PRF-38535 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters	1	1
Final electrical test parameters	1*, 2, 3, 9	1*, 2, 3, 9
Group A test requirements	1,2,3,4**, 9,10,11, (12,13,14)***	1,2,3,4**, 9,10,11, (12,13,14)***
Group B electrical test parameters when using the method 5005 QCI option	1,2,3 and table IV delta limits	N/A
Group C end-point electrical parameters	1,2,3 and table IV delta limits	1 and table IV delta limits
Additional electrical subgroups for group C periodic inspections	N/A	(12,13,14)****
Group D end-point electrical parameters	1,2,3, (12,13,14)*****	1

* PDA applies to subgroup 1.
** Subgroup 4, see 4.4.1c
*** See 4.4.1d
**** See 4.4.3c
***** See 4.4.4

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.4 Technology Conformance inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 5, 6, 7, and 8 shall be omitted.

c. Subgroup 4 (C_A, C_{IS}, and C_{OS} measurements) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and ground at a frequency of 1 MHz. Subgroup 4 shall be performed using a sample of 5 devices with no failures allowed.

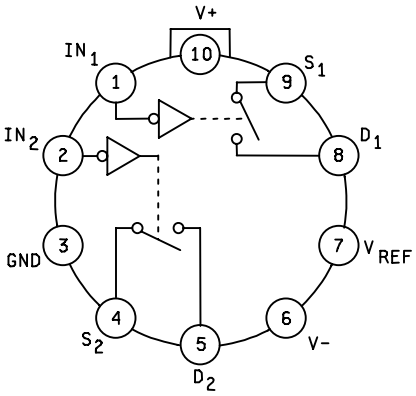
d. Subgroups 12, 13, and 14 shall be added to group A inspection as specified in table III herein. These subgroups shall be performed for initial qualification only using a sample of five devices for each device type submitted to group A inspection, with no failures allowed. If not more than one failure is found in the first sample of five, a second sample of five is permitted with no further failures allowed.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of MIL-PRF-38535.

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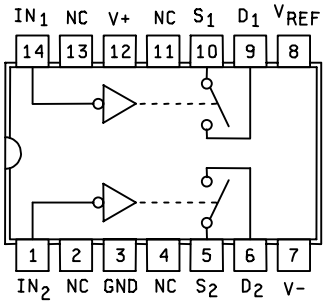
DEVICE TYPE 01

CASE I



DEVICE TYPE 01

CASE C



DEVICE TYPE 02

CASE E

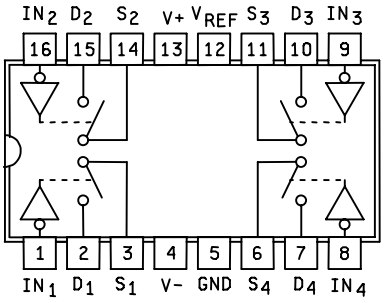


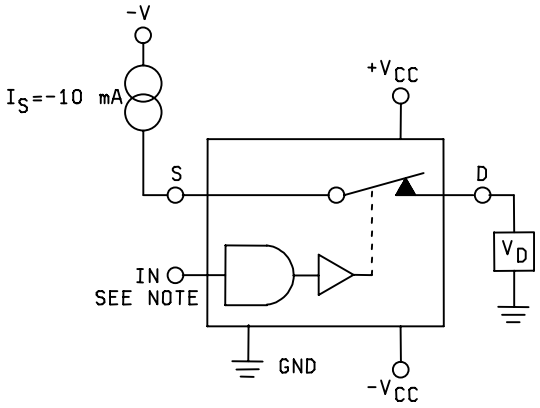
Figure 1. Terminal connections.

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Truth table for each switch	
Input	Switch condition
$2.4\ V \leq V_{IN} \leq 15.0\ V$	OPEN
$0\ V \leq V_{IN} \leq 0.8\ V$	Closed

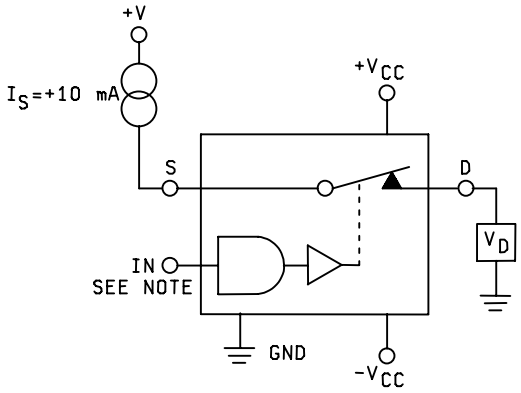
Figure 2. Truth table (all device types).

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NOTE: V_{IN} from table I

$R_{DS} = (V_S - V_D)/-10$ mA

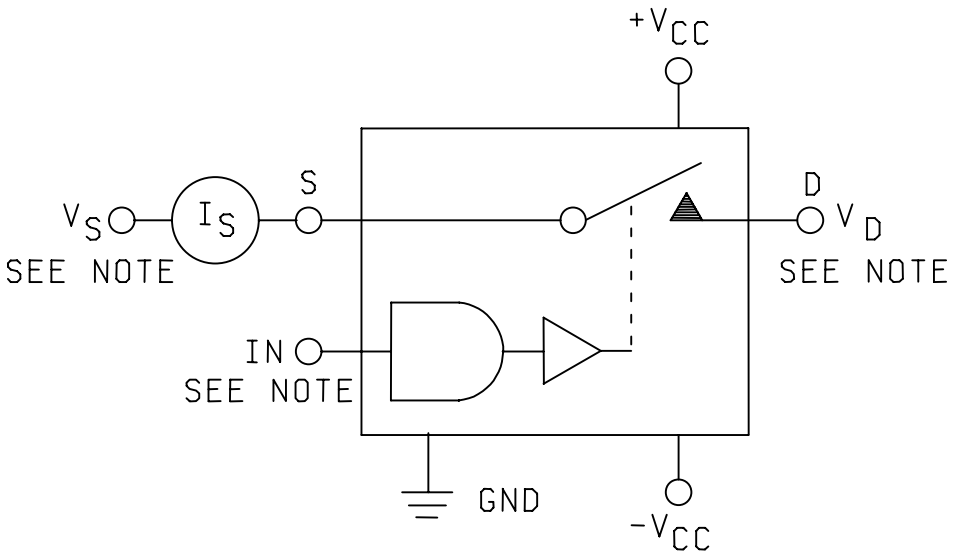


NOTE: V_{IN} from table I

$R_{DS} = (V_S - V_D)/10$ mA

Figure 3. R_{DS} test circuits.

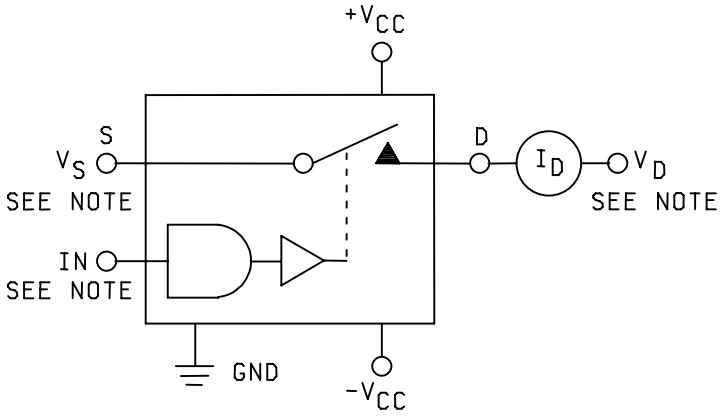
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NOTE: Test conditions are from table I.

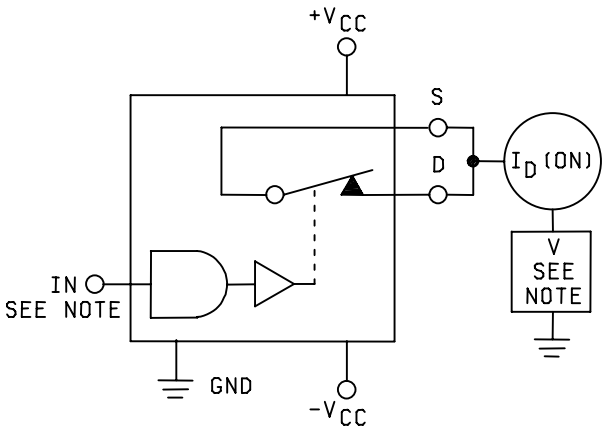
Figure 4. $I_{S(OFF)}$ test circuit.

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NOTE: Test conditions are from table I.

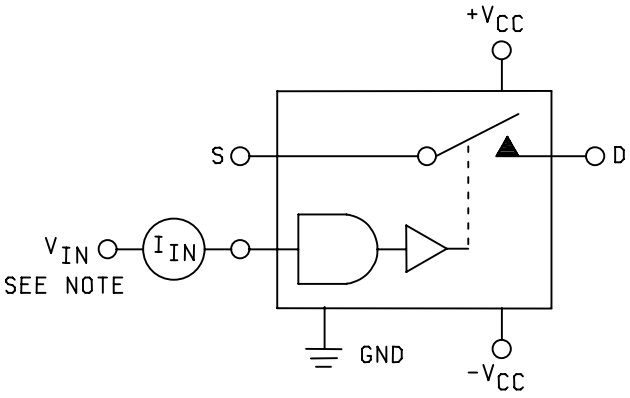
Figure 5. $I_{D(OFF)}$ test circuit.



NOTE: Conditions are from table I.

Figure 6. $I_{D(ON)}$ test circuit.

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NOTE: Test conditions are from table I.

Figure 7. I_{IL} , I_{IH} test circuit.

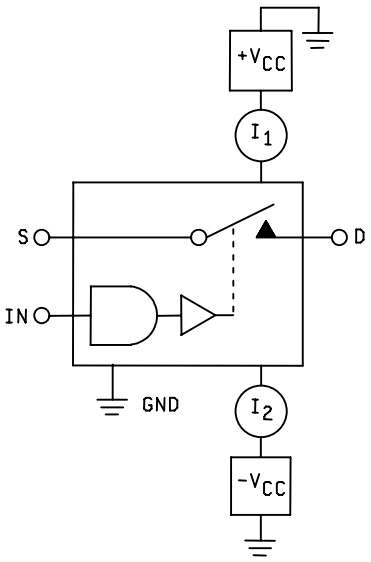
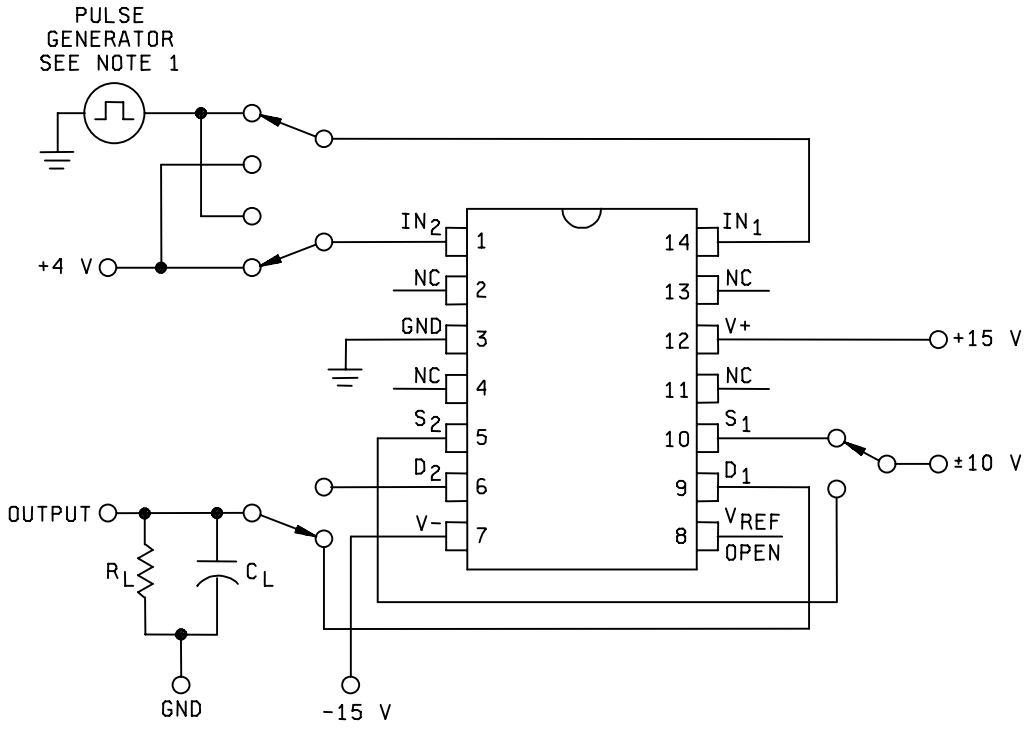


Figure 8. $\pm I_{CC}$ test circuit.

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Device type 01

Case C



TEST CIRCUIT

R_L = 1 kΩ ± 5%
C_L = 100 pF ± 5%
(Includes wiring and probe capacitance.)

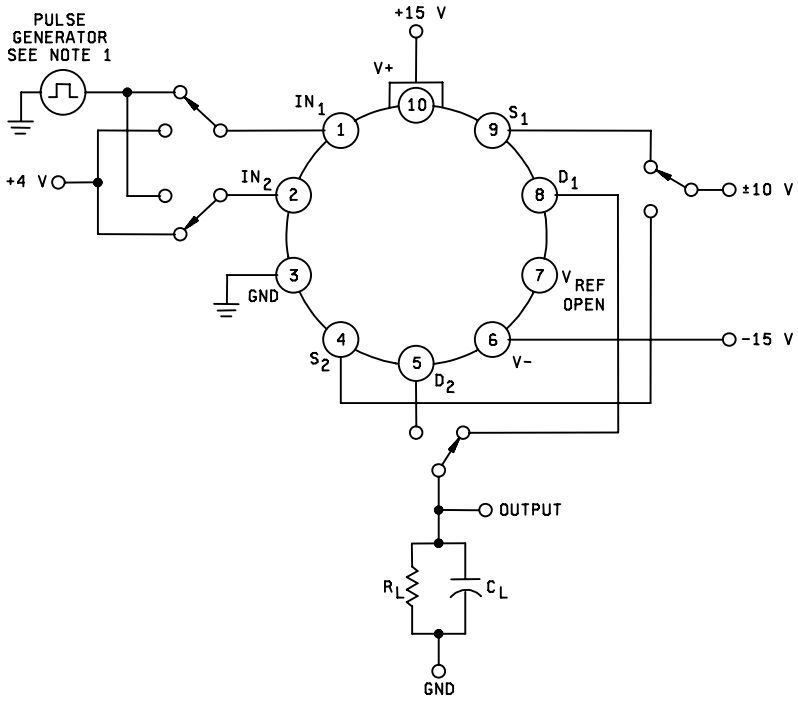
NOTES:
1. The pulse generator has the following characteristics:
   V_{GEN} = 4 V, t_{THL} and $t_{TLH} \leq$ 20 ns.
2. See table III for complete terminal conditions.

Figure 9. Switching time test circuits and waveforms.

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Device type 01

Case I



TEST CIRCUIT

R_L = 1 kΩ ± 5%
C_L = 100 pF ± 5%
(Includes wiring and probe capacitance.)

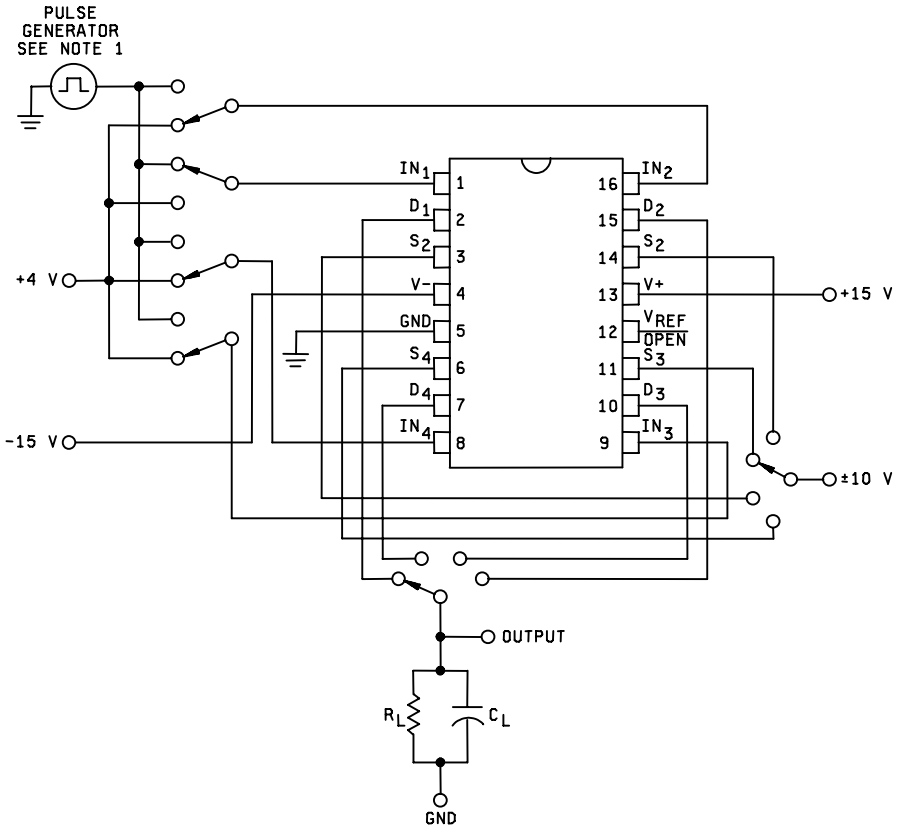
NOTES:
1. The pulse generator has the following characteristics:
   V_{GEN} = 4 V, t_{THL} and $t_{TLH} \leq$ 20 ns.
2. See table III for complete terminal conditions.

FIGURE 9. Switching time test circuits and waveforms – Continued.

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Device type 02

Case E



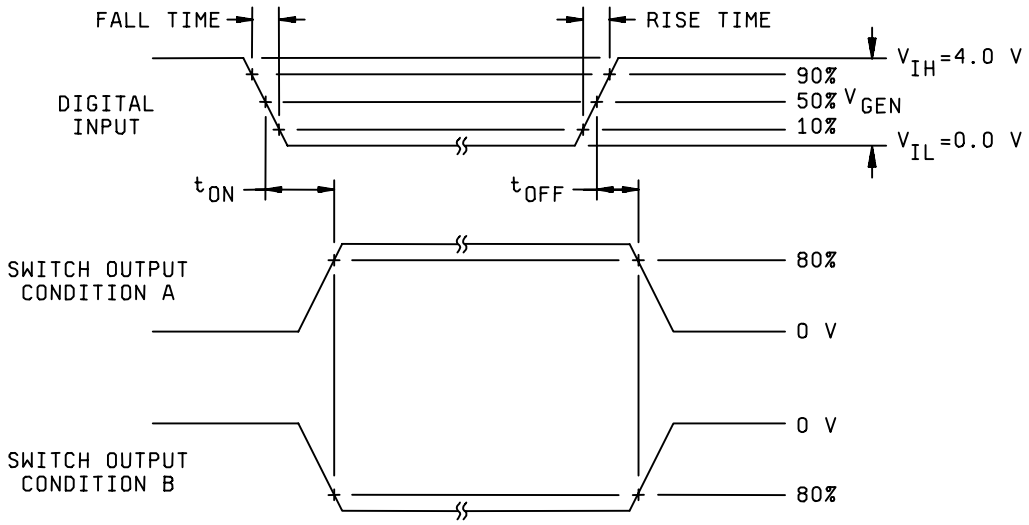
R_L = 1 kΩ ± 5%
C_L = 100 pF ± 5%
(Includes wiring and probe capacitance.)

NOTES:
1. The pulse generator has the following characteristics:
   V_{GEN} = 4 V, $t_{THL} \leq$ 20 ns.
2. See table III for complete terminal conditions.

FIGURE 9. Switching time test circuits and waveforms – Continued.

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Device types 01 and 02

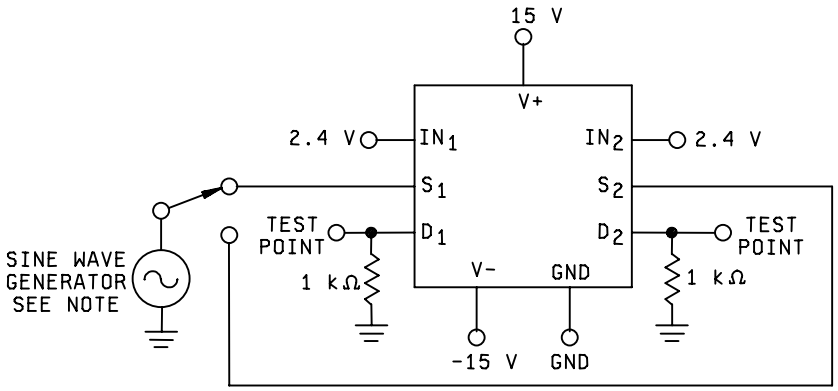


NOTE: Rise time and fall time ≤ 20 ns.

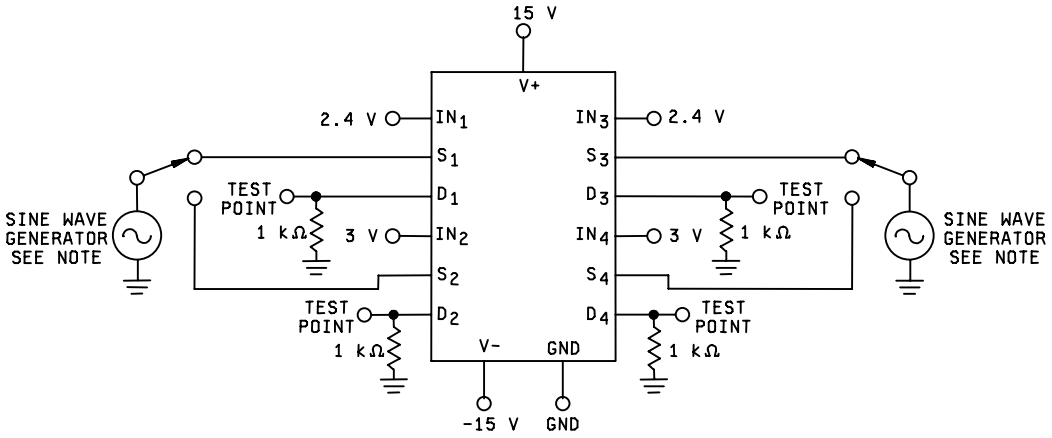
FIGURE 9. Switching time test circuits and waveforms – Continued.

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Device type 01



Device type 02

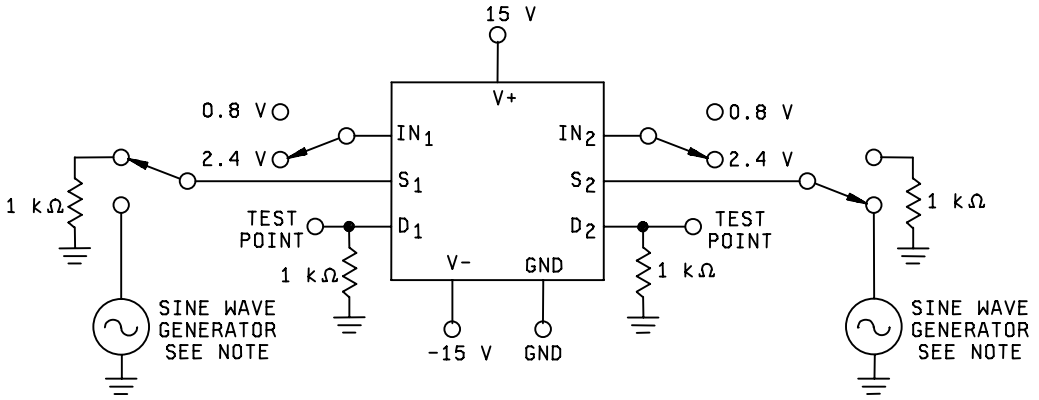


NOTE: The sine wave generator has the following characteristics:
V_{GEN} = 1 V_{P-P}, frequency = 200 kHz.

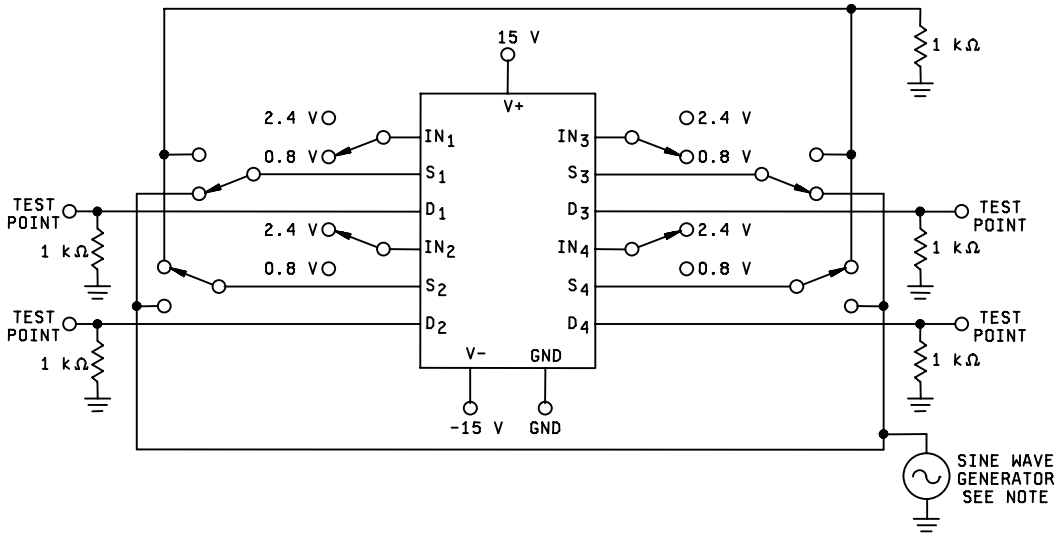
FIGURE 10. Off isolation test circuits.

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Device type 01



Device type 02

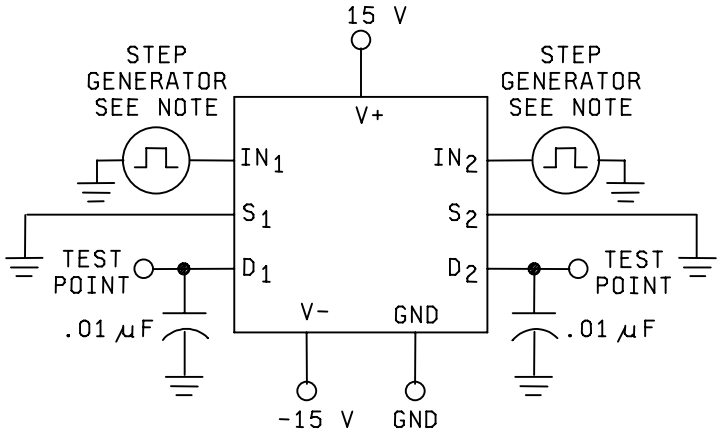


NOTE: The pulse generator has the following characteristics:
V_{GEN} = 1 V_{P-P}, frequency = 200 kHz.

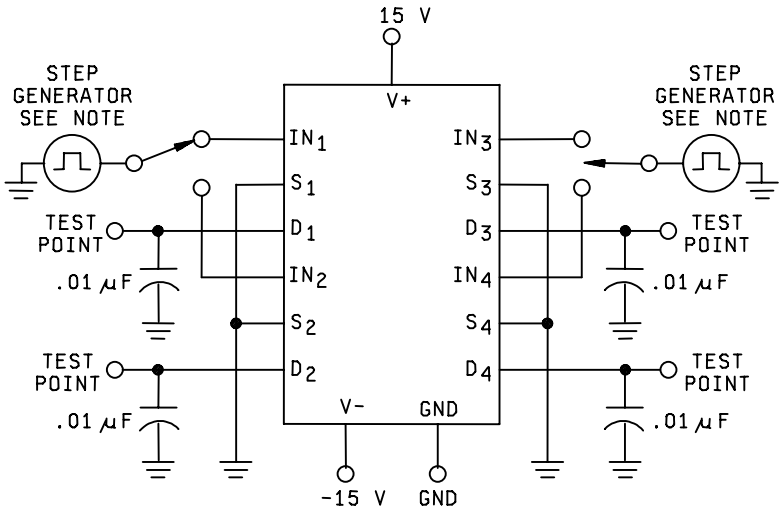
FIGURE 11. Crosstalk test circuits.

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Device type 01



Device type 02



NOTE: The pulse generator has the following characteristics:
V_{GEN} = 0 to 4 V, rise time ≤ 20 ns, fall time ≤ 20 ns, PRR = 200 kHz.

FIGURE 12. Charge transfer error test circuits.

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TABLE III. Group A inspection for device type 01.

Subgroup	Symbol	MIL-STD-883 Method	Case C	Terminal conditions 1/														Measured terminal	Test limits		Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14				
			Test no.	IN_2	NC	GND	NC	S_2	D_2	V-	V_{REF}	D_1	S_1	NC	V+	NC	IN_1		Min	Max	
1 T_A =+25°C	R_{DS}		1	0.8 V		GND				-15 V		-1 mA	10 V		15 V		0.8 V	9-10		70	Ω
			2	"		"		10 V	-1 mA	-15 V					15 V		"	6-5		70	"
			3	"		"				-10 V		-1 mA	7.5 V		10 V		"	9-10		100	"
			4	"		"		7.5 V	-1 mA	-10 V					10 V		"	6-5		100	"
			5	"		"				-15 V		1 mA	-10 V		15 V		"	9-10		70	"
			6	"		"		-10 V	1 mA	-15 V					15 V		"	6-5		70	"
			7	"		"				-10 V		1 mA	-7.5 V		10 V		"	9-10		100	"
			8	"		"		-7.5 V	1 mA	-10 V					10 V		"	6-5		100	"
	$I_{S(OFF)}$		9	2.4 V		"				-15 V		-14 V	14 V		15 V		2.4 V	10	-2	2	nA
			10	"		"		14 V	-14 V	"					"		"	5	"	"	"
			11	"		"				"		14 V	-14 V		"		"	10	"	"	"
			12	"		"		-14 V	14 V	"					"		"	5	"	"	"
	$I_{D(OFF)}$		13	"		"				"		-14 V	14 V		"		"	9	"	"	"
			14	"		"		14 V	-14 V	"					"		"	6	"	"	"
			15	"		"				"		14 V	-14 V		"		"	9	"	"	"
			16	"		"		-14 V	14 V	"					"		"	6	"	"	"
	$I_{D(ON)}$		17	0.8 V		"				"		14 V	14 V		"		0.8 V	9,10	"	"	"
			18	"		"		14 V	14 V	"					"		"	5,6	"	"	"
			19	"		"				"		-14 V	-14 V		"		"	9,10	"	"	"
			20	"		"		-14 V	-14 V	"					"		"	5,6	"	"	"
	I_{IL}	3009	21	2.4 V		"				"					"		"	14	-0.5	0.5	µA
	I_{IL}	3009	22	0.8 V		"				"					"		2.4 V	1	"	"	"
	I_{IH}	3010	23	0.8 V		"				"					"		2.4 V	14	"	"	"
		"	24	2.4 V		"				"					"		0.8 V	1	"	"	"
		"	25	0.8 V		"				"					"		15.0 V	14	"	"	"
		"	26	15.0 V		"				"					"		0.8 V	1	"	"	"
	$+I_{CC}$	3005	27	0.8 V		"				"					"		0.8 V	12		1.5	mA
	$+I_{CC}$	"	28	2.4 V		"				"					"		2.4 V	12		1.5	"
	$-I_{CC}$	"	29	0.8 V		"				"					"		0.8 V	7	-1.5		"
	$-I_{CC}$	"	30	2.4 V		"				"					"		2.4 V	7	-1.5		"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 – Continued.

Subgroup	Symbol	MIL-STD-883 Method	Case C Test no.	Terminal conditions 1/ 1 IN_2	2 NC	3 GND	4 NC	5 S_2	6 D_2	7 V-	8 V_{REF}	9 D_1	10 S_1	11 NC	12 V+	13 NC	14 IN_1	Measured terminal	Test limits Min	Test limits Max	Unit
2 T_A =125°C	R_{DS}		31	0.8 V		GND				-15 V		-1 mA	10 V		15 V		0.8 V	9-10		100	Ω
			32	"		"		10 V	-1 mA	-15 V					15 V		"	6-5		100	"
			33	"		"				-10 V		-1 mA	7.5 V		10 V		"	9-10		150	"
			34	"		"		7.5 V	-1 mA	-10 V					10 V		"	6-5		150	"
			35	"		"				-15 V		1 mA	-10 V		15 V		"	9-10		100	"
			36	"		"		-10 V	1 mA	-15 V					15 V		"	6-5		100	"
			37	"		"				-10 V		1 mA	-7.5 V		10 V		"	9-10		150	"
			38	"		"		-7.5 V	1 mA	-10 V					10 V		"	6-5		150	"
	$I_{S(OFF)}$		39	2.4 V		"				-15 V		-14 V	14 V		15 V		2.4 V	10	-100	100	nA
			40	"		"		14 V	-14 V	"					"		"	5	"	"	"
			41	"		"				"		14 V	-14 V		"		"	10	"	"	"
			42	"		"		-14 V	14 V	"					"		"	5	"	"	"
	$I_{D(OFF)}$		43	"		"				"		-14 V	14 V		"		"	9	"	"	"
			44	"		"		14 V	-14 V	"					"		"	6	"	"	"
			45	"		"				"		14 V	-14 V		"		"	9	"	"	"
			46	"		"		-14 V	14 V	"					"		"	6	"	"	"
	$I_{D(ON)}$		47	0.8 V		"				"		14 V	14 V		"		0.8 V	9,10	-200	200	"
			48	"		"		14 V	14 V	"					"		"	5,6	"	"	"
			49	"		"				"		-14 V	-14 V		"		"	9,10	"	"	"
			50	"		"		-14 V	-14 V	"					"		"	5,6	"	"	"
	I_{IL}	3009	51	2.4 V		"				"					"		"	14	-1.0	1.0	µA
	I_{IL}	3009	52	0.8 V		"				"					"		2.4 V	1	"	"	"
	I_{IH}	3010	53	0.8 V		"				"					"		2.4 V	14	"	"	"
		"	54	2.4 V		"				"					"		0.8 V	1	"	"	"
		"	55	0.8 V		"				"					"		15.0 V	14	"	"	"
		"	56	15.0 V		"				"					"		0.8 V	1	"	"	"
	$+I_{CC}$	3005	57	0.8 V		"				"					"		0.8 V	12		1.5	mA
	$+I_{CC}$	"	58	2.4 V		"				"					"		2.4 V	12		1.5	"
	$-I_{CC}$	"	59	0.8 V		"				"					"		0.8 V	7	-1.5		"
	$-I_{CC}$	"	60	2.4 V		"				"					"		2.4 V	7	-1.5		"
3 T_A =-55°C	R_{DS}		61	0.8 V		"				"		-1 mA	10 V		"		0.8 V	9-10		70	Ω
			62	"		"		10 V	-1 mA	"					"		"	6-5		70	"
			63	"		"				-10 V		-1 mA	7.5 V		10 V		"	9-10		100	"
			64	"		"		7.5 V	-1 mA	-10 V					10 V		"	6-5		100	"
			65	"		"				-15 V		1 mA	-10 V		15 V		"	9-10		70	"
			66	"		"		-10 V	1 mA	-15 V					15 V		"	6-5		70	"
			67	"		"				-10 V		1 mA	-7.5 V		10 V		"	9-10		100	"
			68	"		"		-7.5 V	1 mA	-10 V					10 V		"	6-5		100	"
	$I_{S(OFF)}$		69	2.4 V		"				-15 V		-14 V	14 V		15 V		2.4 V	10	-100	100	nA
			70	"		"		14 V	-14 V	"					"		"	5	"	"	"
			71	"		"				"		14 V	-14 V		"		"	10	"	"	"
			72	"		"		-14 V	14 V	"					"		"	5	"	"	"

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TABLE III. Group A inspection for device type 01 – Continued.

Subgroup	Symbol	MIL-STD-883 Method	Case C	Terminal conditions 1/														Measured terminal	Test limits		Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14				
			Test no.	IN_2	NC	GND	NC	S_2	D_2	V-	V_{REF}	D_1	S_1	NC	V+	NC	IN_1		Min	Max	
3 T_A =-55°C	$I_{D(OFF)}$		73	2.4 V		GND				-15 V		-14 V	14 V		15 V		2.4 V	9	-100	100	nA
			74	"		"		14 V	-14 V	"					"		"	6	"	"	"
			75	"		"				"		14 V	-14 V		"		"	9	"	"	"
			76	"		"		-14 V	14 V	"					"		"	6	"	"	"
	$I_{D(ON)}$		77	0.8 V		"				"		14 V	14 V		"		0.8 V	9,10	-200	200	"
			78	"		"		14 V	14 V	"					"		"	5,6	"	"	"
			79	"		"				"		-14 V	-14 V		"		"	9,10	"	"	"
			80	"		"		-14 V	-14 V	"					"		"	5,6	"	"	"
	I_{IL}	3009	81	2.4 V		"				"					"		"	14	-1.0	1.0	µA
	I_{IL}	3009	82	0.8 V		"				"					"		2.4 V	1	"	"	"
	I_{IH}	3010	83	0.8 V		"				"					"		2.4 V	14	"	"	"
		"	84	2.4 V		"				"					"		0.8 V	1	"	"	"
		"	85	0.8 V		"				"					"		15.0 V	14	"	"	"
		"	86	15.0 V		"				"					"		0.8 V	1	"	"	"
	$+I_{CC}$	3005	87	0.8 V		"				"					"		0.8 V	12		2.0	mA
	$+I_{CC}$	"	88	2.4 V		"				"					"		2.4 V	12		2.0	"
	$-I_{CC}$	"	89	0.8 V		"				"					"		0.8 V	7	-2.0		"
	$-I_{CC}$	"	90	2.4 V		"				"					"		2.4 V	7	-2.0		"
4 T_A = 25°C	C_A 2/	3012	91			"				"					"		2/ 0 V	14		15	pF
	C_A 2/	"	92	2/ 0 V		"				"					"			1		"	"
	C_{IS} 2/	"	93			"				"			2/		"		5 V	10		"	"
	C_{IS} 2/	"	94	5 V		"		2/		"					"			5		"	"
	C_{OS} 3/	"	95			"				"		3/			"		5 V	9		20	"
	C_{OS} 3/	"	96	5 V		"			3/	"					"			6		20	"
9 T_A =25°C	t_{ON}	3003 Fig. 9	97			"				"		OUT	10 V		"		IN	14-9		600	ns
			98	IN		"		10 V	OUT	"					"			1-6		"	"
		"	99			"				"		OUT	-10 V		"		IN	14-9		"	"
		"	100	IN		"		-10 V	OUT	"					"			1-6		"	"
	t_{OFF}	"	101			"				"		OUT	10 V		"		IN	14-9		500	"
		"	102	IN		"		10 V	OUT	"					"			1-6		"	"
		"	103			"				"		OUT	-10 V		"		IN	14-9		"	"
		"	104	IN		"		-10 V	OUT	"					"			1-6		"	"

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TABLE III. Group A inspection for device type 01 – Continued.

Subgroup	Symbol	MIL-STD-883 Method	Case C Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
				IN_2	NC	GND	NC	S_2	D_2	V-	V_{REF}	D_1	S_1	NC	V+	NC	IN_1				
9	t_{ON}	3003	105			GND				-15 V		OUT	10 V		15 V		IN	14-9		800	ns
T_A =25°C		Fig. 9	106	IN		"		10 V	OUT	"					"			1-6		"	"
		"	107			"				"		OUT	-10 V		"		IN	14-9		"	"
		"	108	IN		"		-10 V	OUT	"					"			1-6		"	"
	t_{OFF}	"	109			"				"		OUT	10 V		"		IN	14-9		650	"
		"	110	IN		"		10 V	OUT	"					"			1-6		"	"
		"	111			"				"		OUT	-10 V		"		IN	14-9		"	"
		"	112	IN		"		-10 V	OUT	"					"			1-6		"	"
11	t_{ON}	"	113			"				"		OUT	10 V		"		IN	14-9		600	"
T_A = -55°C		"	114	IN		"		10 V	OUT	"					"			1-6		"	"
		"	115			"				"		OUT	-10 V		"		IN	14-9		"	"
		"	116	IN		"		-10 V	OUT	"					"			1-6		"	"
	t_{OFF}	"	117			"				"		OUT	10 V		"		IN	14-9		500	"
		"	118	IN		"		10 V	OUT	"					"			1-6		"	"
		"	119			"				"		OUT	-10 V		"		IN	14-9		"	"
		"	120	IN		"		-10 V	OUT	"					"			1-6		"	"
12	V_{ISO}	Fig. 10	121	2.4 V		"				"			IN		"		2.4 V	9	60		dB
T_A = 25°C	V_{ISO}	Fig. 10	122	2.4 V		"		IN		"					"		2.4 V	6	"		"
13	V_{CT}	Fig. 11	123	0.8 V		"		IN		"					"		2.4 V	9	60		"
T_A = 25°C																					
14	V_{CTE}	Fig. 12	124	2.4 V		"				"					"		IN	9	-10	10	mV
T_A = 25°C		Fig. 12	125	IN		"		GND		"			GND		"		2.4 V	6	-10	10	mV

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TABLE III. Group A inspection for device type 01 – Continued.

Subgroup	Symbol	MIL-STD-883 Method	Case I	Terminal conditions 1/										Measured terminal	Test limits		Unit
				1	2	3	4	5	6	7	8	9	10				
			Test no.	IN_1	IN_2	GND	S_2	D_2	V-	V_{REF}	D_1	S_1	V+		Min	Max	
1 T_A = 25°C	R_{DS}		1	0.8 V	0.8 V	GND			-15 V		-1 mA	10 V	15 V	9-8		70	Ω
			2	"	"	"	10 V	-1 mA	-15 V				15 V	4-5		70	"
			3	"	"	"			-10 V		-1 mA	7.5 V	10 V	9-8		100	"
			4	"	"	"	7.5 V	-1 mA	-10 V				10 V	4-5		100	"
			5	"	"	"			-15 V		1 mA	-10 V	15 V	8-9		70	"
			6	"	"	"	-10 V	1 mA	-15 V				15 V	5-4		70	"
			7	"	"	"			-10 V		1 mA	-7.5 V	10 V	8-9		100	"
			8	"	"	"	-7.5 V	1 mA	-10 V				10 V	5-4		100	"
	$I_{S(OFF)}$		9	2.4 V	2.4 V	"			-15 V		-14 V	14 V	15 V	9	-2	2	nA
			10	"	"	"	14 V	-14 V	"				"	4	"	"	"
			11	"	"	"			"		14 V	-14 V	"	9	"	"	"
			12	"	"	"	-14 V	14 V	"				"	4	"	"	"
	$I_{D(OFF)}$		13	"	"	"			"		14 V	-14 V	"	8	"	"	"
			14	"	"	"	-14 V	14 V	"				"	5	"	"	"
			15	"	"	"			"		-14 V	14 V	"	8	"	"	"
			16	"	"	"	14 V	-14 V	"				"	5	"	"	"
	$I_{D(ON)}$		17	0.8 V	0.8 V	"			"		14 V	14 V	"	8	"	"	"
			18	"	"	"	14 V	14 V	"				"	5	"	"	"
			19	"	"	"			"		-14 V	-14 V	"	8	"	"	"
			20	"	"	"	-14 V	-14 V	"				"	5	"	"	"
	I_{IL}	3009	21	"	2.4 V	"			"				"	1	-0.5	0.5	µA
	I_{IL}	3009	22	2.4 V	0.8 V	"			"				"	2	"	"	"
	I_{IH}		23	2.4 V	0.8 V	"			"				"	1	"	"	"
			24	0.8 V	2.4 V	"			"				"	2	"	"	"
			25	15.0 V	0.8 V	"			"				"	1	"	"	"
			26	0.8 V	15.0 V	"			"				"	2	"	"	"
	$+I_{CC}$	3005	27	0.8 V	0.8 V	"			"				"	10		1.5	mA
	$+I_{CC}$	"	28	2.4 V	2.4 V	"			"				"	10		1.5	"
	$-I_{CC}$	"	29	0.8 V	0.8 V	"			"				"	6	-1.5		"
	$-I_{CC}$	"	30	2.4 V	2.4 V	"			"				"	6	-1.5		"
2 T_A = 125°C	R_{DS}		31	0.8 V	0.8 V	"			"		-1 mA	10 V	"	9-8		100	Ω
			32	"	"	"	10 V	-1 mA	"				"	4-5		100	"
			33	"	"	"			-10 V		-1 mA	7.5 V	10 V	9-8		150	"
			34	"	"	"	7.5 V	-1 mA	-10 V				10 V	4-5		150	"
			35	"	"	"			-15 V		1 mA	-10 V	15 V	9-8		100	"
			36	"	"	"	-10 V	1 mA	-15 V				15 V	4-5		100	"
			37	"	"	"			-10 V		1 mA	-7.5 V	10 V	9-8		150	"
			38	"	"	"	-7.5 V	1 mA	-10 V				10 V	4-5		150	"
	$I_{S(OFF)}$		39	2.4 V	2.4 V	"			-15 V		-14 V	14 V	15 V	9	-100	100	nA
			40	"	"	"	14 V	-14 V	"				"	4	"	"	"
			41	"	"	"			"		14 V	-14 V	"	9	"	"	"
			42	"	"	"	-14 V	14 V	"				"	4	"	"	"
	$I_{D(OFF)}$		43	"	"	"			"		-14 V	14 V	"	8	"	"	"
			44	"	"	"	14 V	-14 V	"				"	5	"	"	"
			45	"	"	"			"		14 V	-14 V	"	8	"	"	"
			46	"	"	"	-14 V	14 V	"				"	5	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 – Continued.

Subgroup	Symbol	MIL-STD-883 Method	Case I	Terminal conditions 1/										Measured terminal	Test limits		Unit
				1	2	3	4	5	6	7	8	9	10				
			Test no.	IN_1	IN_2	GND	S_2	D_2	V-	V_{REF}	D_1	S_1	V+		Min	Max	
2 T_A = 125°C	$I_{D(ON)}$		47	0.8 V	0.8 V	GND			-15 V		14 V	14 V	15 V	8	-200	200	nA
			48	"	"	"	14 V	14 V	"				"	5	"	"	"
			49	"	"	"			"		-14 V	-14 V	"	8	"	"	"
			50	"	"	"	-14 V	-14 V	"				"	5	"	"	"
	I_{IL}	3009	51	"	2.4 V	"			"				"	1	-1.0	1.0	µA
	I_{IL}	3009	52	2.4 V	0.8 V	"			"				"	2	"	"	"
	I_{IH}	3010	53	2.4 V	0.8 V	"			"				"	1	"	"	"
	"	"	54	0.8 V	2.4 V	"			"				"	2	"	"	"
	"	"	55	15.0 V	0.8 V	"			"				"	1	"	"	"
	"	"	56	0.8 V	15.0 V	"			"				"	2	"	"	"
	$+I_{CC}$	3005	57	0.8 V	0.8 V	"			"				"	10		1.5	mA
	$+I_{CC}$	"	58	2.4 V	2.4 V	"			"				"	10		1.5	"
	$-I_{CC}$	"	59	0.8 V	0.8 V	"			"				"	6	-1.5		"
	$-I_{CC}$	"	60	2.4 V	2.4 V	"			"				"	6	-1.5		"
3 T_A = -55°C	R_{DS}		61	0.8 V	0.8 V	"			"		-1 mA	10 V	"	9-8		70	Ω
			62	"	"	"	10 V	-1 mA	"				"	4-5		70	"
			63	"	"	"			-10 V		-1 mA	7.5 V	10 V	9-8		100	"
			64	"	"	"	7.5 V	-1 mA	-10 V				10 V	4-5		100	"
			65	"	"	"			-15 V		1 mA	-10 V	15 V	9-8		70	"
			66	"	"	"	-10 V	1 mA	-15 V				15 V	4-5		70	"
			67	"	"	"			-10 V		1 mA	-7.5 V	10 V	9-8		100	"
			68	"	"	"	-7.5 V	1 mA	-10 V				10 V	4-5		100	"
	$I_{S(OFF)}$		69	2.4 V	2.4 V	"			-15 V		-14 V	14 V	15 V	9	-100	100	nA
			70	"	"	"	14 V	-14 V	"				"	4	"	"	"
			71	"	"	"			"		14 V	-14 V	"	9	"	"	"
			72	"	"	"	-14 V	14 V	"				"	4	"	"	"
	$I_{D(OFF)}$		73	"	"	"			"		-14 V	14 V	"	8	"	"	"
			74	"	"	"	14 V	-14 V	"				"	5	"	"	"
			75	"	"	"			"		14 V	-14 V	"	8	"	"	"
			76	"	"	"	-14 V	14 V	"				"	5	"	"	"
	$I_{D(ON)}$		77	0.8 V	0.8 V	"			"		14 V	14 V	"	8	-200	200	"
			78	"	"	"	14 V	14 V	"				"	5	"	"	"
			79	"	"	"			"		-14 V	-14 V	"	8	"	"	"
			80	"	"	"	-14 V	-14 V	"				"	5	"	"	"
	I_{IL}	3009	81	"	2.4 V	"			"				"	1	-1.0	1.0	µA
	I_{IL}	3009	82	2.4 V	0.8 V	"			"				"	2	"	"	"
	I_{IH}	3010	83	2.4 V	0.8 V	"			"				"	1	"	"	"
	"	"	84	0.8 V	2.4 V	"			"				"	2	"	"	"
	"	"	85	15.0 V	0.8 V	"			"				"	1	"	"	"
	"	"	86	0.8 V	15.0 V	"			"				"	2	"	"	"
	$+I_{CC}$	3005	87	0.8 V	0.8 V	"			"				"	10		2.0	mA
	$+I_{CC}$	"	88	2.4 V	2.4 V	"			"				"	10		2.0	"
	$-I_{CC}$	"	89	0.8 V	0.8 V	"			"				"	6	-2.0		"
	$-I_{CC}$	"	90	2.4 V	2.4 V	"			"				"	6	-2.0		"

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TABLE III. Group A inspection for device type 01 – Continued.

Subgroup	Symbol	MIL-STD-883 Method	Case I	Terminal conditions 1/										Measured terminal	Test limits		Unit
				1	2	3	4	5	6	7	8	9	10				
			Test no.	IN_1	IN_2	GND	S_2	D_2	V-	V_{REF}	D_1	S_1	V+		Min	Max	
4 T_A = 25°C	C_A 2/ C_A	3012 “	91 92	2/ 0 V	2/ 0 V	GND “			-15 V “				15 V “	1 2		15 “	pF “
	C_{IS} 2/ C_{IS}	“ “	93 94	5 V	5 V	“ “	2/		“ “			2/	“ “	9 4		“ “	“ “
	C_{OS} 3/ C_{OS} 3/	“ “	95 96	5 V	5 V	“ “		3/	“ “		3/		“ “	8 5		20 20	“ “
9 T_A = 25°C	t_{ON}	3003 Fig. 9	97 98	IN	IN	“ “	10 V	OUT	“ “		OUT	10 V	“ “	1-8 2-5		600 “	ns “
		“ “	99 100	IN	IN	“ “	-10 V	OUT	“ “		OUT	-10 V	“ “	1-8 2-5		“ “	“ “
	t_{OFF}	“ “	101 102	IN	IN	“ “	10 V	OUT	“ “		OUT	10 V	“ “	1-8 2-5		500 “	“ “
		“ “	103 104	IN	IN	“ “	-10 V	OUT	“ “		OUT	-10 V	“ “	1-8 2-5		“ “	“ “
10 T_A = 125°C	t_{ON}	“ “	105 106	IN	IN	“ “	10 V	OUT	“ “		OUT	10 V	“ “	1-8 2-5		800 “	“ “
		“ “	107 108	IN	IN	“ “	-10 V	OUT	“ “		OUT	-10 V	“ “	1-8 2-5		“ “	“ “
	t_{OFF}	“ “	109 110	IN	IN	“ “	10 V	OUT	“ “		OUT	10 V	“ “	1-8 2-5		650 “	“ “
		“ “	111 112	IN	IN	“ “	-10 V	OUT	“ “		OUT	-10 V	“ “	1-8 2-5		“ “	“ “
11 T_A = -55°C	t_{ON}	“ “	113 114	IN	IN	“ “	10 V	OUT	“ “		OUT	10 V	“ “	1-8 2-5		600 “	“ “
		“ “	115 116	IN	IN	“ “	-10 V	OUT	“ “		OUT	-10 V	“ “	1-8 2-5		“ “	“ “
	t_{OFF}	“ “	117 118	IN	IN	“ “	10 V	OUT	“ “		OUT	10 V	“ “	1-8 2-5		500 “	“ “
		“ “	119 120	IN	IN	“ “	-10 V	OUT	“ “		OUT	-10 V	“ “	1-8 2-5		“ “	“ “
12 T_A = 25°C	V_{ISO} V_{ISO}	Fig. 10 Fig. 10	121 122	2.4 V “	2.4 V “	“ “	IN		“ “			IN	“ “	8 5	60 “		dB “
13 T_A = 25°C	V_{CT}	Fig. 11	123	“	“	“			“				“	8	“		“
14 T_A = 25°C	V_{CTE}	Fig. 12	124 125	IN 2.4 V	“ IN	“ “	GND		“ “			GND	“ “	5 8	-10 -10	10 10	mV mV

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TABLE III. Group A inspection for device type 02.

Subgroup	Symbol	MIL-STD-883 Method	Case E	Terminal conditions 1/																Measured terminal	Test limits		Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	
			Test no.	IN_1	D_1	S_1	V-	GND	S_4	D_4	IN_4	IN_3	D_3	S_3	V_{REF}	V+	S_2	D_2	IN_2				
1 T_A =+25°C	R_{DS}		1	0.8 V	-1 mA	10 V	-15 V	GND			0.8 V	0.8 V				15 V			0.8 V	3-2		175	Ω
			2	"			"	"			"	"				"	10 V	-1 mA	"	14-15		"	"
			3	"			"	"			"	"	-1 mA	10 V		"			"	11-10		"	"
			4	"			"	"	10 V	-1 mA	"	"				"			"	6-7		"	"
			5	"	-1 mA	7.5 V	-10 V	"			"	"				10 V			"	3-2		200	"
			6	"			"	"			"	"				"	7.5 V	-1 mA	"	14-15		"	"
			7	"			"	"			"	"	-1 mA	7.5 V		"			"	11-10		"	"
			8	"			"	"	7.5 V	-1 mA	"	"				"			"	6-7		"	"
			9	"	1 mA	-10 V	-15 V	"			"	"				15 V			"	2-3		175	"
			10	"			"	"			"	"				"	-10 V	1 mA	"	15-14		"	"
			11	"			"	"			"	"	1 mA	-10 V		"			"	10-11		"	"
			12	"			"	"	-10 V	1 mA	"	"				"			"	7-6		"	"
			13	"	1 mA	-7.5 V	-10 V	"			"	"				10 V			"	2-3		200	"
			14	"			"	"			"	"				"	-7.5 V	1 mA	"	15-14		"	"
			15	"			"	"			"	"	1 mA	-7.5 V		"			"	10-11		"	"
			16	"			"	"	-7.5 V	1 mA	"	"				"			"	7-6		"	"
	$I_{S(OFF)}$		17	2.4 V	-14 V	14 V	-15 V	"			2.4 V	2.4 V				15 V			2.4 V	3	-2	2	nA
			18	"			"	"			"	"				"	14 V	14 V	"	14	"	"	"
			19	"			"	"			"	"	-14 V	14 V		"			"	11	"	"	"
			20	"			"	"	14 V	-14 V	"	"				"			"	6	"	"	"
			21	"	14 V	-14 V	"	"			"	"				"	-14 V	-14 V	"	3	"	"	"
			22	"			"	"			"	"				"			"	14	"	"	"
			23	"			"	"			"	"	14 V	-14 V		"			"	11	"	"	"
			24	"			"	"	-14 V	14 V	"	"				"			"	6	"	"	"
	$I_{D(OFF)}$		25	"	-14 V	14 V	"	"			"	"				"			"	2	"	"	"
			26	"			"	"			"	"				"	14 V	-14 V	"	15	"	"	"
			27	"			"	"			"	"	-14 V	14 V		"			"	10	"	"	"
			28	"			"	"	14 V	-14 V	"	"				"			"	7	"	"	"
			29	"	14 V	-14 V	"	"			"	"				"	-14 V	14 V	"	2	"	"	"
			30	"			"	"			"	"				"			"	15	"	"	"
			31	"			"	"			"	"	14 V	-14 V		"			"	10	"	"	"
			32	"			"	"	-14 V	14 V	"	"				"			"	7	"	"	"
	$I_{D(ON)}$		33	0.8 V	14 V	14 V	"	"			0.8 V	0.8 V				"			0.8 V	2	"	"	"
			34	"			"	"			"	"				"	14 V	14 V	"	15	"	"	"
			35	"			"	"			"	"	14 V	14 V		"			"	10	"	"	"
			36	"			"	"	14 V	14 V	"	"				"			"	7	"	"	"
			37	"	-14 V	-14 V	"	"			"	"				"	-14 V	-14 V	"	2	"	"	"
			38	"			"	"			"	"				"			"	15	"	"	"
			39	"			"	"			"	"	-14 V	-14 V		"			"	10	"	"	"
			40	"			"	"	-14 V	-14 V	"	"				"			"	7	"	"	"
	I_{IL}	3009	41	"			"	"			2.4 V	2.4 V				"			2.4 V	1	-0.5	0.5	µA
		"	42	2.4 V			"	"			"	2.4 V				"			0.8 V	16	"	"	"
		"	43	"			"	"			"	0.8 V				"			2.4 V	9	"	"	"
		"	44	"			"	"			0.8 V	2.4 V				"			2.4 V	8	"	"	"

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TABLE III. Group A inspection for device type 02 – Continued.

Subgroup	Symbol	MIL-STD-883 Method	Case E	Terminal conditions 1/															Measured terminal	Test limits		Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	
			Test no.	IN_1	D_1	S_1	V-	GND	S_4	D_4	IN_4	IN_3	D_3	S_3	V_{REF}	V+	S_2	D_2	IN_2				
1	I_{IH}	3010	45	2.4 V			-15 V	GND			0.8 V	0.8 V				15 V			0.8 V	1	-0.5	0.5	µA
T_A = 25°C		"	46	0.8 V			"	"			"	0.8 V				"			2.4 V	16	"	"	"
		"	47	"			"	"			"	2.4 V				"			0.8 V	9	"	"	"
		"	48	"			"	"			2.4 V	0.8 V				"			0.8 V	8	"	"	"
		"	49	15.0 V			"	"			0.8 V	0.8 V				"			0.8 V	1	"	"	"
		"	50	0.8 V			"	"			"	0.8 V				"			15.0 V	16	"	"	"
		"	51	"			"	"			"	15.0 V				"			0.8 V	9	"	"	"
		"	52	"			"	"			15.0 V	0.8 V				"			0.8 V	8	"	"	"
	$+I_{CC}$	3005	53	"			"	"			0.8 V	0.8 V				"			0.8 V	13		1.5	mA
	$+I_{CC}$	"	54	2.4 V			"	"			2.4 V	2.4 V				"			2.4 V	13		1.5	"
	$-I_{CC}$	"	55	0.8 V			"	"			0.8 V	0.8 V				"			0.8 V	4	-1.5		"
	$-I_{CC}$	"	56	2.4 V			"	"			2.4 V	2.4 V				"			2.4 V	4	-1.5		"
2	R_{DS}		57	0.8 V	-1 mA	10 V	"	"			0.8 V	0.8 V				"			0.8 V	3-2		250	Ω
T_A =125°C			58	"			"	"			"	"				"	10 V	-1 mA	"	14-15		"	"
			59	"			"	"			"	"	-1 mA	10 V		"			"	11-10		"	"
			60	"			"	"	10 V	-1 mA	"	"				"			"	6-7		"	"
			61	"	-1 mA	7.5 V	-10 V	"			"	"				10 V			"	3-2		"	"
			62	"			"	"			"	"				"	7.5 V	-1 mA	"	14-15		"	"
			63	"			"	"			"	"	-1 mA	7.5 V		"			"	11-10		"	"
			64	"			"	"	7.5 V	-1 mA	"	"				"			"	6-7		"	"
			65	"	1 mA	-10 V	-15 V	"			"	"				15 V			"	2-3		"	"
			66	"			"	"			"	"				"	-10 V	1 mA	"	15-14		"	"
			67	"			"	"			"	"	1 mA	-10 V		"			"	10-11		"	"
			68	"			"	"	-10 V	1 mA	"	"				"			"	7-6		"	"
			69	"	1 mA	-7.5 V	-10 V	"			"	"				10 V			"	2-3		"	"
			70	"			"	"			"	"				"	-7.5 V	1 mA	"	15-14		"	"
			71	"			"	"			"	"	1 mA	-7.5 V		"			"	10-11		"	"
			72	"			"	"	-7.5 V	1 mA	"	"				"			"	7-6		"	"
	$I_{S(OFF)}$		73	2.4 V	-14 V	14 V	-15 V	"			2.4 V	2.4 V				15 V			2.4 V	3	-100	"	nA
			74	"			"	"			"	"				"	14 V	-14 V	"	14	"	"	"
			75	"			"	"			"	"	-14 V	14 V		"			"	11	"	"	"
			76	"			"	"	14 V	-14 V	"	"				"			"	6	"	"	"
			77	"	14 V	-14 V	"	"			"	"				"			"	3	"	"	"
			78	"			"	"			"	"				"	-14 V	14 V	"	14	"	"	"
			79	"			"	"			"	"	14 V	-14 V		"			"	11	"	"	"
			80	"			"	"	-14 V	14 V	"	"				"			"	6	"	"	"
	$I_{D(OFF)}$		81	"	-14 V	14 V	"	"			"	"				"			"	2	"	"	"
			82	"			"	"			"	"				"	14 V	-14 V	"	15	"	"	"
			83	"			"	"			"	"	-14 V	14 V		"			"	10	"	"	"
			84	"			"	"	14 V	-14 V	"	"				"			"	7	"	"	"
			85	0.8 V	14 V	-14 V	"	"			0.8 V	0.8 V				"			0.8 V	2	"	"	"
			86	"			"	"			"	"				"	-14 V	14 V	"	15	"	"	"
			87	"			"	"			"	"	14 V	-14 V		"			"	10	"	"	"
			88	"			"	"	-14 V	14 V	"	"				"			"	7	"	"	"
	$I_{D(ON)}$		89	"	14 V	14 V	"	"			"	"				"			"	2	-200	200	"
			90	"			"	"			"	"				"	14 V	14 V	"	15	"	"	"
			91	"			"	"			"	"	14 V	14 V		"			"	10	"	"	"
			92	"			"	"	14 V	14 V	"	"				"			"	7	"	"	"
			93	"	-14 V	-14 V	"	"			"	"				"			"	2	"	"	"
			94	"			"	"			"	"				"	-14 V	-14 V	"	15	"	"	"
			95	"			"	"			"	"	-14 V	-14 V		"			"	10	"	"	"
			96	"			"	"	-14 V	-14 V	"	"				"			"	7	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 – Continued.

Subgroup	Symbol	MIL-STD-883 Method	Case E	Terminal conditions 1/															Measured terminal	Test limits		Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	
			Test no.	IN_1	D_1	S_1	V-	GND	S_4	D_4	IN_4	IN_3	D_3	S_3	V_{REF}	V+	S_2	D_2	IN_2				
2	I_{IL}	3009	97	0.8 V			-15 V	GND			2.4 V	2.4 V				15 V			2.4 V	1	-1.0	1.0	µA
T_A =125°C		"	98	2.4 V			"	"			"	2.4 V				"			0.8 V	16	"	"	"
		"	99	"			"	"			"	0.8 V				"			2.4 V	9	"	"	"
		"	100	"			"	"			0.8 V	2.4 V				"			2.4 V	8	"	"	"
	I_{IH}	3010	101	"			"	"			"	0.8 V				"			0.8 V	1	"	"	"
		"	102	0.8 V			"	"			"	0.8 V				"			2.4 V	16	"	"	"
		"	103	0.8 V			"	"			"	2.4 V				"			"	9	"	"	"
		"	104	0.8 V			"	"			2.4 V	0.8 V				"			"	8	"	"	"
		"	105	15.0 V			"	"			0.8 V	"				"			0.8 V	1	"	"	"
		"	106	0.8 V			"	"			"	"				"			15.0 V	16	"	"	"
		"	107	"			"	"			"	15.0 V				"			0.8 V	9	"	"	"
		"	108	"			"	"			15.0 V	0.8 V				"			"	8	"	"	"
	$+I_{CC}$	3005	109	"			"	"			0.8 V	0.8 V				"			"	13		1.5	mA
	$+I_{CC}$	"	110	2.4 V			"	"			2.4 V	2.4 V				"			2.4 V	13		1.5	"
	$-I_{CC}$	"	111	0.8 V			"	"			0.8 V	0.8 V				"			0.8 V	4	-1.5		"
	$-I_{CC}$	"	112	2.4 V			"	"			2.4 V	2.4 V				"			2.4 V	4	-1.5		"
3	R_{DS}		113	0.8 V	-1 mA	10 V	"	"			0.8 V	0.8 V				"			0.8 V	3-2		175	Ω
T_A = -55°C			114	"			"	"			"	"				"	10 V	-1 mA	"	14-15		"	"
			115	"			"	"			"	"	-1 mA	10 V		"			"	11-10		"	"
			116	"			"	"	10 V	-1 mA	"	"				"			"	6-7		"	"
			117	"	-1 mA	7.5 V	-10 V	"			"	"				10 V			"	3-2		200	"
			118	"			"	"			"	"				"	7.5 V	-1 mA	"	14-15		"	"
			119	"			"	"			"	"	-1 mA	7.5 V		"			"	11-10		"	"
			120	"			"	"	7.5 V	-1 mA	"	"				"			"	6-7		"	"
			121	"	1 mA	-10 V	-15 V	"			"	"				15 V			"	2-3		175	"
			122	"			"	"			"	"				"	-10 V	1 mA	"	15-14		"	"
			123	"			"	"			"	"	1 mA	-10 V		"			"	10-11		"	"
			124	"			"	"	-10 V	1 mA	"	"				"			"	7-6		"	"
			125	"	1 mA	-7.5 V	-10 V	"			"	"				10 V			"	2-3		200	"
			126	"			"	"			"	"				"	-7.5 V	1 mA	"	15-14		"	"
			127	"			"	"			"	"	1 mA	-7.5 V		"			"	10-11		"	"
			128	"			"	"	-7.5 V	1 mA	"	"				"			"	7-6		"	"
	$I_{S(OFF)}$		129	2.4 V	-14 V	14 V	-15 V	"			2.4 V	2.4 V				15 V			2.4 V	3	-100	100	nA
			130	"			"	"			"	"				"	14 V	-14 V	"	14	"	"	"
			131	"			"	"			"	"	-14 V	14 V		"			"	11	"	"	"
			132	"			"	"	14 V	-14 V	"	"				"			"	6	"	"	"
			133	"	14 V	-14 V	"	"			"	"				"			"	3	"	"	"
			134	"			"	"			"	"				"	-14 V	14 V	"	14	"	"	"
			135	"			"	"			"	"	14 V	-14 V		"			"	11	"	"	"
			136	"			"	"	-14 V	14 V	"	"				"			"	6	"	"	"
	$I_{D(OFF)}$		137	"	-14 V	14 V	"	"			"	"				"			"	2	"	"	"
			138	"			"	"			"	"				"	14 V	-14 V	"	15	"	"	"
			139	"			"	"			"	"	-14 V	14 V		"			"	10	"	"	"
			140	"			"	"	14 V	-14 V	"	"				"			"	7	"	"	"
			141	"	14 V	-14 V	"	"			"	"				"			"	2	"	"	"
			142	"			"	"			"	"				"	-14 V	14 V	"	15	"	"	"
			143	"			"	"			"	"	14 V	-14 V		"			"	10	"	"	"
			144	"			"	"	-14 V	14 V	"	"				"			"	7	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 – Continued.

Subgroup	Symbol	MIL-STD-883 Method	Case E	Terminal conditions 1/																Measured terminal	Test limits		Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	IN_1	D_1	S_1	V-	GND	S_4	D_4	IN_4	IN_3	D_3	S_3	V_{REF}	V+	S_2	D_2	IN_2		Min	Max	
3 T_A = -55°C	$I_{D(ON)}$	3009	145	0.8 V	14 V	14 V	-15 V	GND			0.8 V	0.8 V				15 V			0.8 V	2	-200	200	nA
	"	"	146	"			"	"			"	"				"	14 V	14 V	"	15	"	"	"
	"	"	147	"			"	"			"	"	14 V	14 V		"			"	10	"	"	"
	"	"	148	"			"	"	14 V	14 V	"	"				"			"	7	"	"	"
	"	"	149	"	-14 V	-14 V	"	"			"	"				"			"	2	"	"	"
	"	"	150	"			"	"			"	"				"	-14 V	-14 V	"	15	"	"	"
	"	"	151	"			"	"			"	"	-14 V	-14 V		"			"	10	"	"	"
	"	"	152	"			"	"	-14 V	-14 V	"	"				"			"	7	"	"	"
	I_{IL}	"	153	"			"	"			2.4 V	2.4 V				"			2.4 V	1	-1.0	1.0	µA
	"	"	154	2.4 V			"	"			"	2.4 V				"			0.8 V	16	"	"	"
	"	"	155	"			"	"			"	0.8 V				"			2.4 V	9	"	"	"
	"	"	156	"			"	"			0.8 V	2.4 V				"			2.4 V	8	"	"	"
	I_{IH}	3010	157	"			"	"			"	0.8 V				"			0.8 V	1	"	"	"
	"	"	158	0.8 V			"	"			"	0.8 V				"			2.4 V	16	"	"	"
	"	"	159	"			"	"			"	2.4 V				"			0.8 V	9	"	"	"
	"	"	160	"			"	"			2.4 V	0.8 V				"			"	8	"	"	"
	"	"	161	15.0 V			"	"			0.8 V	"				"			"	1	"	"	"
	"	"	162	0.8 V			"	"			"	"				"			15.0 V	16	"	"	"
	"	"	163	"			"	"			"	15.0 V				"			0.8 V	9	"	"	"
	"	"	164	"			"	"			15.0 V	0.8 V				"			0.8 V	8	"	"	"
	$+I_{CC}$	3005	165	"			"	"			0.8 V	0.8 V				"			0.8 V	13		2.0	mA
	$+I_{CC}$	"	166	2.4 V			"	"			2.4 V	2.4 V				"			2.4 V	13		2.0	"
	$-I_{CC}$	"	167	0.8 V			"	"			0.8 V	0.8 V				"			0.8 V	4	-2.0		"
	$-I_{CC}$	"	168	2.4 V			"	"			2.4 V	2.4 V				"			2.4 V	4	-2.0		"
4 T_A = 25°C	C_A 2/	3012	169	2/ 0 V			"	"								"				1		15	pF
	"	"	170				"	"								"			2/ 0 V	16		"	"
	"	"	171				"	"				2/ 0 V				"				9		"	"
	"	"	172				"	"			2/ 0 V					"				8		"	"
	C_{IS} 2/	"	173	2/ 5 V			"	"								"				3		"	pF
	"	"	174				"	"								"			2/ 5 V	14		"	"
	"	"	175				"	"				2/ 5 V				"				11		"	"
	"	"	176				"	"			2/ 5 V					"				6		"	"
	C_{OS} 3/	"	177	3/ 5 V			"	"								"				2		20	"
	"	"	178				"	"								"			3/ 5 V	15		"	"
	"	"	179				"	"				3/ 5 V				"				10		"	"
	"	"	180				"	"			3/ 5 V					"				7		"	"
9 T_A = 25°C	t_{ON}	3003 Fig. 9	181	IN	OUT	10 V	"	"								"				1-2		600	ns
	"	"	182				"	"								"	10 V	OUT	IN	16-15		"	"
	"	"	183				"	"				IN	OUT	10 V		"				9-10		"	"
	"	"	184				"	"	10 V	OUT	IN					"				8-7		"	"
	"	"	185	IN	OUT	-10 V	"	"								"				1-2		"	"
	"	"	186				"	"								"	-10 V	OUT	IN	16-15		"	"
	"	"	187				"	"				IN	OUT	-10 V		"				9-10		"	"
	"	"	188				"	"	-10 V	OUT	IN					"				8-7		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 – Continued.

Subgroup	Symbol	MIL-STD-883 Method	Case E	Terminal conditions 1/															Measured terminal	Test limits		Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	
			Test no.	IN_1	D_1	S_1	V-	GND	S_4	D_4	IN_4	IN_3	D_3	S_3	V_{REF}	V+	S_2	D_2	IN_2				
9	t_{OFF}	3003	189	IN	OUT	10 V	-15 V	GND								15 V				1-2		500	ns
T_A = 25°C		Fig. 9	190	"	"	"	"	"								"	10 V	OUT	IN	16-15		"	"
		"	191				"	"				IN	OUT	10 V		"				9-10		"	"
		"	192				"	"	10 V	OUT	IN					"				8-7		"	"
		"	193	IN	OUT	-10 V	"	"								"				1-2		"	"
		"	194				"	"								"	-10 V	OUT	IN	16-15		"	"
		"	195				"	"				IN	OUT	-10 V		"				9-10		"	"
		"	196				"	"	-10 V	OUT	IN					"				8-7		"	"
10	t_{ON}	"	197	IN	OUT	10 V	"	"								"				1-2		800	"
T_A =125°C		"	198				"	"								"	10 V	OUT	IN	16-15		"	"
		"	199				"	"				IN	OUT	10 V		"				9-10		"	"
		"	200				"	"	10 V	OUT	IN					"				8-7		"	"
		"	201	IN	OUT	-10 V	"	"								"				1-2		"	"
		"	202				"	"								"	-10 V	OUT	IN	16-15		"	"
		"	203				"	"				IN	OUT	-10 V		"				9-10		"	"
		"	204				"	"	-10 V	OUT	IN					"				8-7		"	"
	t_{OFF}	"	205	IN	OUT	10 V	"	"								"				1-2		650	"
		"	206				"	"								"	10 V	OUT	IN	16-15		"	"
		"	207				"	"				IN	OUT	10 V		"				9-10		"	"
		"	208				"	"	10 V	OUT	IN					"				8-7		"	"
		"	209	IN	OUT	-10 V	"	"								"				1-2		"	"
		"	210				"	"								"	-10 V	OUT	IN	16-15		"	"
		"	211				"	"				IN	OUT	-10 V		"				9-10		"	"
		"	212				"	"	-10 V	OUT	IN					"				8-7		"	"
11	t_{ON}	"	213	IN	OUT	10 V	"	"								"				1-2		600	"
T_A =-55°C		"	214				"	"								"	10 V	OUT	IN	16-15		"	"
		"	215				"	"				IN	OUT	10 V		"				9-10		"	"
		"	216				"	"	10 V	OUT	IN					"				8-7		"	"
		"	217	IN	OUT	-10 V	"	"								"				1-2		"	"
		"	218				"	"								"	-10 V	OUT	IN	16-15		"	"
		"	219				"	"				IN	OUT	-10 V		"				9-10		"	"
		"	220				"	"	-10 V	OUT	IN					"				8-7		"	"
	t_{OFF}	"	221	IN	OUT	10 V	"	"								"				1-2		500	"
		"	222				"	"								"	10 V	OUT	IN	16-15		"	"
		"	223				"	"				IN	OUT	10 V		"				9-10		"	"
		"	224				"	"	10 V	OUT	IN					"				8-7		"	"
		"	225	IN	OUT	-10 V	"	"								"				1-2		"	"
		"	226				"	"								"	-10 V	OUT	IN	16-15		"	"
		"	227				"	"				IN	OUT	-10 V		"				9-10		"	"
		"	228				"	"	-10 V	OUT	IN					"				8-7		"	"
12	V_{ISO}	Fig. 10	229	2.4 V		IN	"	"			2.4 V	2.4 V				"			2.4 V	2	60		dB
T_A = 25°C		"	230	"			"	"			"	"				"		IN	"	15	"		"
		"	231	"			"	"			"	"		IN		"			"	10	"		"
		"	232	"			"	"	IN		"	"				"			"	7	"		"
13	V_{CT}	Fig. 11	233	2.4 V			"	"	"		0.8 V	0.8 V		IN		"	IN		0.8 V	2	"		"
T_A = 25°C		"	234	0.8 V		IN	"	"	"		"	0.8 V		IN		"			2.4 V	15	"		"
		"	235	"		"	"	"	"		"	2.4 V				"	IN		0.8 V	10	"		"
		"	236	"		"	"	"			2.4 V	0.8 V		IN		"	IN		0.8 V	7	"		"
14	V_{CTE}	Fig. 12	237	IN		GND	"	"			"	2.4 V				"			2.4 V	2	-10	10	mV
T_A = 25°C		"	238	2.4 V			"	"			"	2.4 V				"	GND		IN	15	"		"
		"	239	"			"	"			"	IN		GND		"			2.4 V	10	"		"
		"	240	"			"	"	GND		IN	2.4 V				"			2.4 V	7	"		"

1/ Pins not designated may be "high" level logic, "low" level logic, or open.
2/ Connect capacitance bridge between measured input terminal and GND, f = 1 MHz; see 4.4.1c.
3/ Connect capacitance bridge between measured output terminal and GND, f = 1 MHz; see 4.4.1c.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End point electrical parameters shall be as specified in table II herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

c. A special subgroup shall be added to group C inspection for class B devices only, and it shall consist of subgroups 12, 13, and 14 of group A inspection as specified in table III herein. This special subgroup shall be performed on each device type that is qualified from those listed in 1.2.1 herein. After initial qualification, the special subgroup shall be performed periodically on a single device type selected from those device types previously qualified. A sample of 5 devices (of the device type to be inspected) shall be chosen and submitted to test with no failures allowed. If not more than 1 failure is found in the first sample of 5, a second sample of 5 is permitted with no further failures allowed. When more than one device type is qualified, the single device type selected shall be different device type for each subsequent periodic inspection until all qualified device types have been inspected. The sequence of single device types shall be repeated to fulfill the periodic inspection requirement.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End point electrical parameters shall be as specified in table II herein. A special subgroup shall be added to group D inspection for class S devices only, and it shall consist of subgroups 12, 13, and 14 of group A inspection as specified in table III herein. This special subgroup shall be performed on each device type that is qualified from those listed in 1.2.1 herein. After initial qualification, the special subgroup shall be performed periodically on a single device type selected from those device types previously qualified. When more than one device type is qualified, the single device type selected shall be different device type for each subsequent periodic inspection until all qualified device types have been inspected. The sequence of single device types shall be repeated to fulfill the periodic inspection requirement.

Table IV. Delta limits at 25°C. 1/

Parameter 2/	Device types	
	01	02
R_{DS}	15 Ω	15 Ω
$I_{D(ON)}$	1 nA	1 nA

1/ Each test applies to every switch in the package.
2/ Each of the above parameters shall be recorded before and after the required life tests to determine deltas (Δ).

4.5 Methods of inspection. Methods of inspection shall be specified and as follows.

4.5.1 Voltage and current. All voltage values given are referenced to the microcircuit ground terminals. Currents given are conventional current and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. PIN and compliance identifier, if applicable (see 1.2).

c. Requirements for delivery of one copy of the conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to acquiring activity in addition to notification of the qualifying activity, if applicable.

f. Requirements for failure analysis (including required test condition of MIL-STD-883, method 5003), corrective action and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements should not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

i. Requirements for “JAN” marking.

j. Packaging requirements (see 5.1).

6.3 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractor's parts lists.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, 3990 E. Broad Street, Columbus, Ohio 43123-1199.

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6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535, MIL-HDBK-1331, and as follows:.

V+	Positive supply voltage
V-	Negative supply voltage
V_{REF}	Reference voltage
V_{IL}	Input (address) low voltage
V_{IH}	Input (address) high voltage
I_{CC+}	Positive supply current
I_{CC-}	Negative supply current
$I_{S(OFF)}$	Leakage current into the source terminal of an “OFF” switch
$I_{D(OFF)}$	Leakage current into the drain terminal of an “OFF” switch
$I_{D(ON)}$	Leakage current from an “ON” driver into the switch
I_{IL}	Low level (address) input current
I_{IH}	High level (address) input current
R_{DS}	Resistance of an “ON” switch
GND	Ground zero voltage potential
S	Source terminal to the switch
D	Drain terminal of the switch
V_{ZAP}	Input test voltage
IN	Address input to the switch
T_A	Ambient temperature
V_{CTE}	Charge transfer error
V_{CT}	Crosstalk between channels
V_{ISO}	Off isolation from source to drain of a closed switch

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming should not affect the part number.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

Military device type	Generic-industry type
01	200
02	201

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army – CR
Navy - EC
Air Force - 11
NASA – NA
DLA – CC

Preparing activity:
DLA - CC

Project 5962-2015

Review activities:
Army – MI, SM
Navy – AS, CG, MC, SH, TD
Air Force – 03, 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:27Z
Check the source to verify that this is the current version before use.